

Part 1: Custom Shell Input Handling (C Program)

```
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

int main()
{
    struct termios oldt, newt;
    char buffer[100];
    int index = 0;
    char ch;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1)
    {
        printf("ossh> ");
        fflush(stdout);
        index = 0;
        while (1)
        {
            ch = getchar();
            if (ch == '
' || ch == ')
            {
                buffer[index] = '\0';
                printf("
");
                break;
            }
            else if (ch == 127 || ch == 8)
            {
                if (index > 0)
                {
                    index--;
                    printf(" ");
                    fflush(stdout);
                }
            }
            else if (index < 99)
            {
                buffer[index++] = ch;
                putchar(ch);
                fflush(stdout);
            }
        }

        if (strcmp(buffer, "exit") == 0)
        {
            printf("Exiting shell...
");
            break;
        }

        if (index > 0)
        {
```

```

        printf("You entered: %s
", buffer);
    }
}

    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
    return 0;
}

```

Terminal Output - Skill 2 Execution

```

rishi@RISHI-VP:~/oslab/2520090131_Skill/Skill2$ ./input
ossh> hello
You entered: hello
ossh> ls -l
You entered: ls -l
ossh> hello
You entered: hello
ossh> hell
You entered: hell
ossh> hello
You entered: hello
ossh> hell
You entered: hell
ossh> exit
Exiting shell...

```

Part 2: MiniShell Implementation (C Program)

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    while (1)
    {
        printf("MiniShell> ");
        scanf("%s", command);
        if (strcmp(command, "exit") == 0)
            break;

        pid_t pid = fork();
        if (pid == 0)
        {
            printf("Child PID : %d
", getpid());
            execlp(command, command, NULL);
            printf("Command not found!
");
            exit(1);
        }
        else
        {
            printf("Parent PID : %d
", getpid());
            wait(NULL);
        }
    }
}

```

```
}  
    return 0;  
}
```

Terminal Output - MiniShell Execution

```
rishi@RISHI-VP:~$ cd oslab  
rishi@RISHI-VP:~/oslab$ mkdir minishell  
rishi@RISHI-VP:~/oslab$ cd minishell  
rishi@RISHI-VP:~/oslab/minishell$ nano minishell.c  
rishi@RISHI-VP:~/oslab/minishell$ gcc minishell.c -o minishell  
rishi@RISHI-VP:~/oslab/minishell$ ./minishell  
MiniShell> pwd  
Parent PID : 12868  
Child PID : 12873  
/home/rishi/oslab/minishell  
MiniShell> ls  
Parent PID : 12868  
Child PID : 12874  
minishell minishell.c  
MiniShell> date  
Parent PID : 12868  
Child PID : 12875  
Sat Aug 1 04:06:42 UTC 2026  
MiniShell> exit  
rishi@RISHI-VP:~/oslab/minishell$
```